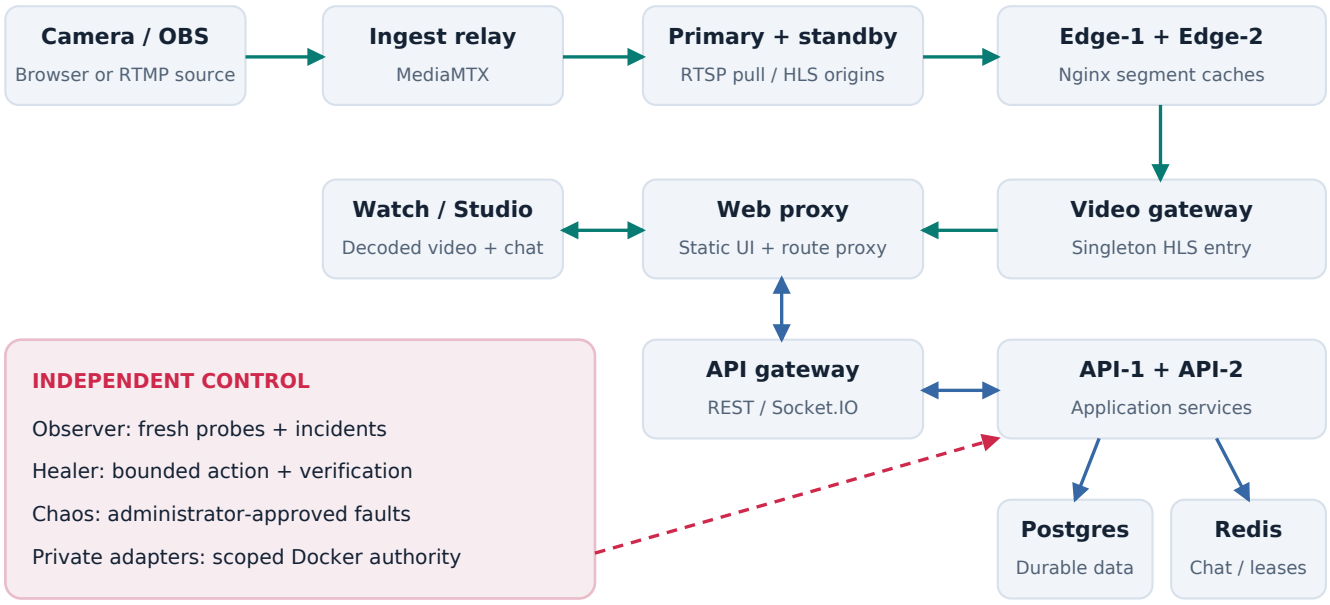
ENGINEERING REPORT / RECORDED EXPERIMENTS

StreamShield Resilience report

Real streaming, controlled failures and independently verified restoration.

4 / 4	13	513
faults verified	containers cleaned	unit checks passed

Architecture and responsibility boundaries



Green: media Blue: application Dashed red: probes / scoped actions

Logical flow; API-2 and control roles share one fixture container. One host.

Creators publish via WHIP/WebRTC or RTMP. MediaMTX relays and remuxes to primary/standby origins; Nginx edges and the video gateway deliver HLS. Express serves accounts, metadata and Socket.IO chat, while PostgreSQL stores durable records and Redis distributes ephemeral state.

Scope and test environment

Run **a484d151** on 27 October 2026: Linux Docker 28.4.0, 5 reported CPUs, 33.3 GiB reported memory, vfs storage, Chromium and one generated H.264/AAC live path. These are daemon-reported resources, not a peak-usage measurement. The compact fixture omits Prometheus, Grafana and API-3.

Persistent application: **127.0.0.1:5173**. Disposable streaming/chaos demo: **127.0.0.1:5174**. They use separate projects and data; demo faults exclude the persistent application. Physical Windows/8 GB laptop performance is not established by this Linux run.

RESILIENCE / 02

Failure detection and healing

Each crash uses actual SIGKILL in a newly owned fixture. Network disruption uses real Toxiproxy latency on a simulated read-only recommendation path. The manual scenario is separate historical acceptance evidence.

Failure	How detected	How restored / verified	Recovery time
API server crash	Independent readiness probe becomes unreachable; consecutive failures confirm the incident. **Detect: 7.684 s**	API-2 serves requests. Healer starts API-1; readiness, catalog and gateway participation verify recovery.	**21.500 s** Verified
Video gateway crash	Gateway health is unreachable; real HLS delivery through it fails. **Detect: 6.497 s**	Healer starts the gateway; valid playlists/init/media and advancing segments through it verify recovery.	**15.324 s** Verified
Edge node failure	Edge-1 probe fails; actual delivery is observed through Edge-2. **Detect: 4.700 s**	Edge-2 carries delivery. Healer starts Edge-1 and independently verifies live media; peer remains unchanged.	**14.232 s** Verified
Network disruption	Safe recommendation reads time out; the dependency circuit opens and the observer confirms failure. **Detect: 9.055 s**	Owned 20-second delay expires/is removed. Circuit half-open reads verify reconnection; this is not a third-party repair.	**27.247 s** Verified
Media authorization loss	Both video routes deny new HLS; buffered playback drains and stops.	Operator restores simulated signing authority; advancing video must then be verified. No real key is destroyed or recreated.	**Not recorded** Human action required

What the recovery number means

Recovery time = **independent service verification timestamp - fault injection timestamp**. It includes detection, action and verification. Docker action duration and total demonstration duration are separate. A 15-second observation target is not a guaranteed recovery deadline.

For API and edge failures, redundancy may preserve video. For gateway failure, a buffered player may continue briefly before stalling. Network-fault time includes the injected delay's expiry and successful circuit reads. The fifth scenario's duration was not stored; no estimate is substituted.

Current result

All four automated experiments returned their affected cards to Healthy with retained recovery timestamps, real playback progress and verified service behavior. The three crash cases recorded **docker_start_accepted**; the network case recorded safe-read/circuit reconnection evidence. The control-plane container remained running.

RESILIENCE / 03

Before-and-after numbers

The first table tracks service state across the same experiment. The running flags are binary process observations, not throughput or audience-capacity measurements.

Signal	Before fault	During fault	After verification
API-1 running flag	1	0; exit 137	1; functional checks pass
Video gateway running flag	1	0; exit 137	1; HLS progresses
Edge-1 running flag	1	0; Edge-2 serves	1; peer unchanged
Recommendation read	2 ms	Timeout / circuit opens	1 ms; circuit closed

Actual decoded playback after each restoration

Restored scenario	Decoded frames	Media-time advance	New media responses
API server crash	792 to 809 (+17)	+0.701 s	2
Video gateway crash	183 to 200 (+17)	+0.700 s	2
Edge node failure	727 to 757 (+30)	+1.242 s	2
Network disruption	1565 to 1596 (+31)	+1.268 s	2

These are short post-restoration samples of a 640 x 360 player. They prove resumed/continuing playback, not total outage or buffering duration. Playback interruption was unmeasured (null) in this run; null does not mean zero.

Comparison with the earlier recorded local run

Scenario	Earlier 282d20da	Current a484d151	Change
API server crash	16.503 s	21.500 s	+4.997 s
Video gateway crash	13.259 s	15.324 s	+2.065 s
Edge node failure	17.383 s	14.232 s	-3.151 s
Network disruption	26.227 s	27.247 s	+1.020 s

Earlier: 26 October; current: 27 October. This is a descriptive cross-run comparison, not a controlled benchmark: host load and code differ. Both slower and faster results are retained. No causal speedup, percentile or reliability improvement is inferred from two local runs.

Control policy and reproducibility

Recovery is an evidence-driven state machine

Fresh failures are confirmed by the Observer. The Healer selects a component/symptom playbook, checks dependencies and authority, persists intent, requests an allowed action, and independently verifies stable behavior. SCHEDULED → EXECUTING → VERIFYING → VERIFIED; unsupported, ambiguous or exhausted work escalates.

Guard	Enforced behavior
Identity and authority	Administrator + CSRF/origin checks + typed target; adapter identifies the exact owned Compose project and service.
Durable actions	Local SQLite intent/lease epochs and PostgreSQL audit records; ambiguous actions are not blindly replayed.
Bounded repair	Default 2 operations/service/10 minutes; 30-second cooldown. Never restart the last serving API or mutate PostgreSQL automatically.
Fresh functional proof	Readiness/catalog/gateway participation for API; real initialization/segment bytes and progression for media. Stale or missing evidence stays unknown.

Run the submitted chaos script

Use Node.js 24, Docker with Linux containers and Playwright Chromium. Prepare the pinned images and configuration using README.md first. Use one fixture at a time.

```
npm ci
npm run setup:platform
npm run build
npx playwright install chromium

npm run chaos -- --help
npm run chaos -- --verify
npm run chaos -- --interactive
```

The new entry point delegates to the existing isolated driver in the same Node process, preserving Windows Ctrl+C cleanup. It accepts only the documented modes; arbitrary target/host arguments are rejected before Docker is touched. Interactive controls expose the fifth, operator-only case.

Evidence and cleanup

Fresh automated run: **4/4** verified scenarios; **13** own containers removed; local user environment unchanged. Unit/security suite: **513** checks across **44** files, plus typecheck and lint. The real fixture compiled current services/static assets and decoded video.

Detailed results: test-results/light-four/runs/a484d151/measurements.json. A safe selected snapshot is committed as docs/RESILIENCE_REPORT_DATA.json. Request IDs, timestamps and receipts correlate evidence; private keys and passwords are not report data.

RESILIENCE / 05

Known limits and evidence register

One host and singleton services. All redundant peers share one Docker host. Host loss is not healed. Database, relay, gateways, Redis and web remain singleton boundaries.

Compact privilege boundary. API-2 and trusted control processes share a container. The restricted API peer is denied Docker access, but privileged adapters still require host trust; this is not multi-tenant production isolation.

Source and external authority. Only the broadcaster can restore missing camera/capture/internet input. Lost private keys or external signing permissions require authorized human restoration. The fifth failure simulates that limit; it does not integrate a real key vault.

Timed cleanup is not infrastructure repair. Toxiproxy affects a private simulated dependency. The network fault expires and circuit reads reconnect; no claim is made about repairing an ISP or external provider.

Measurements are bounded observations. One generated source and one cloud-host run do not establish production capacity or tail latency. The 50,000-viewer target remains unverified; no zero-interruption claim is made.

Windows and resource limits. Physical devices, VPN/firewall behavior and an 8 GB Windows laptop are not certified here. Local startup delays were observed on that PC. The larger Linux host's results are not transferable performance guarantees.

Public deployment work. Local HTTP/RTMP and loopback ICE need a deliberate TLS, STUN/TURN/NAT, secrets, off-host backup, independent failure-domain and delivery-capacity design before public deployment.

Evidence register / repository navigation

[1] Current run and selected raw numbers — docs/RESILIENCE_REPORT_DATA.json

[2] Earlier four-fault evidence: 282d20da — docs/RECOVERY_HEALTH_VALIDATION.json

[3] Separate manual authority-loss acceptance — docs/SYSTEM_LOGS_VALIDATION.json

[4] Browser-startup / real recovery regression — docs/BROWSER_READINESS_VALIDATION.json

[5] README and reproducible setup — README.md

[6] Submitted chaos entry point — scripts/chaos.mjs

[7] Actual experiment driver — scripts/four-scenario-live.mjs

Application baseline: main commit **48cf25b** (same application tree as a783fdb), plus the submitted chaos entry point and report. This report records local verification; it does not assert that remote GitHub Actions jobs or physical Windows checks passed.